AE's comments

1. My apologies, but I maintain that your H_0 (page 3) is unnatural and uninteresting. In the trend analysis of multiple time series, it is extremely unlikely that the natural 'position of ignorance' is that all trend curves are the same. What if, for example, I already know or suspect that the curves form distinct groups? It is then something of a wasted effort to start from the position that they do not differ at all. Also, the null hypothesis is not robust to high-dimensionality as it is more likely that some curves will differ as I observe a higher number of variates (I understand that you do not do this in the paper but this only makes it even less interesting, in my view).
   As a cartoon illustration of the first point above, if I suspect that p-1 (say) curves may be the same, but 1 is distinctly different, do I need to first remove the offending one before performing your test, in which case I enter the territory of post-selection inference (which you do not cover), or do I not need to remove the offending one, in which case I am pretending that my H_0 is different from what it is in reality?
2. The clustering part is nice and perhaps it could form the main thread of a much shorter paper.
3. The empirical application is interesting, but this is thanks to the use of clustering rather than as a consequence of testing - the null is obviously inappropriate, as Figure 6 suggests, so there is no point in assuming it only to surely reject it. In other words, "let us test our null of equality between all trends" doesn't seem a scientifically valid suggestion for this dataset. The suggestion of clustering the countries is fine, though.

Less major comments

1. Can your model (1.1) include autoregression? Either way, it would be good to see it spelled out explicitly.
2. I was unconvinced by the subtraction of the overall level on p. 6. Firstly, this clearly risks eliminating useful information (what if some curves differ from others in the overall level)? Secondly, I am not sure if it is meaningful to look at averages of (e.g. increasing) trends - for example, what is the interpretation of a "long-term average house price level in [a] country"? In "Time Series 101", we teach students not to take averages of non-stationary time series!
3. (C5) - you use the delta notation before defining it. Why do your covariates X have to be random, e.g. why can't you have one of them as (say) a linear trend?
4. Your comment (iii) on p. 9. But isn't it the case that for near-unit root processes, you will have a variety of different scenarios (some stochastic trends pointing strongly upwards, some downwards, some with no specific direction) - a situation that should be easy to discriminate from your typical scenarios of interest?
5. P. 11. "Throughout the paper, we assume that the long-run error variance is the same for all i, (...) [but we] use a different estimator of σ^2 for each i" - this seems suboptimal.
6. I was disappointed to see that the method was not able to handle missing values.